

Engineering Statistics

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Chapter 15: Statistical Quality Control

Why Quality Matters

- ▶ Quality determines competitiveness.
- ▶ Poor quality increases costs.
- ▶ Customers expect reliable products.
- ▶ Quality management improves processes.

Quality of Design

- ▶ Design quality concerns product features, performance, materials, and functionality.
- ▶ It is driven by customer expectations and willingness to pay.
- ▶ Examples: paper cup vs. crystal goblet; pickup truck vs. luxury sedan.
- ▶ Good design quality matches the needs of target customers.

Quality of Conformance

- ▶ Conformance quality measures how consistently outputs meet design specifications.
- ▶ Good design alone does not prevent manufacturing defects.
- ▶ Process control and monitoring help maintain consistency.
- ▶ High quality requires both good design and conformance.

Deming's Philosophy of Quality

- ▶ Leadership must take responsibility for quality improvement.
- ▶ Prevent defects rather than relying on inspection.
- ▶ Improve systems continuously using data and learning.
- ▶ Reduce fear and encourage cooperation and reporting.

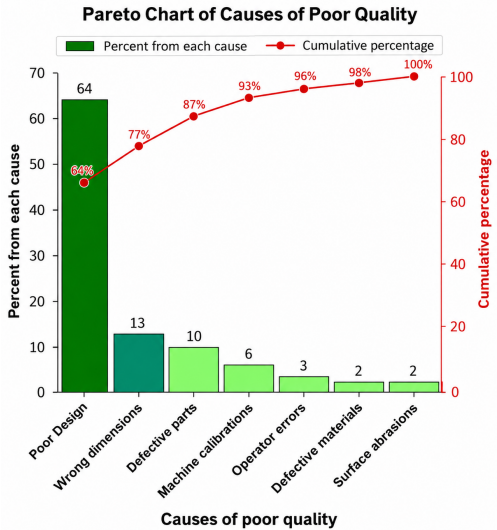
Six Sigma

- ▶ Developed at Motorola in the 1980s to reduce defects.
- ▶ Focuses on measurable financial and quality improvements.
- ▶ Uses structured projects with defined objectives and responsibilities.
- ▶ **DMAIC:** Define, Measure, Analyze, Improve, Control.

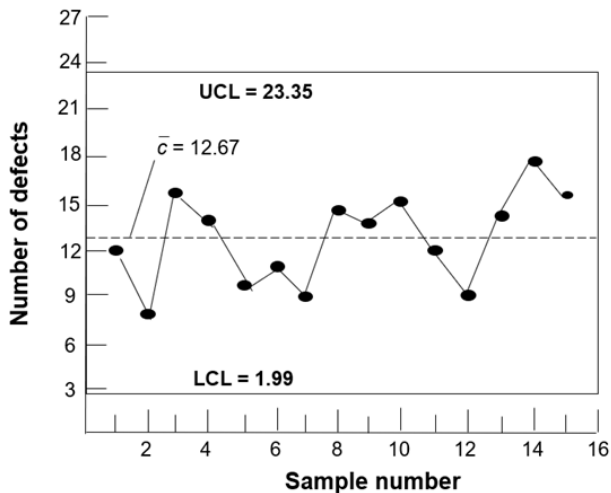
Seven Quality Control Tools

- ▶ Pareto charts and flowcharts.
- ▶ Check sheets and histograms.
- ▶ Scatter diagrams and control charts.
- ▶ Fishbone (cause–effect) diagrams.

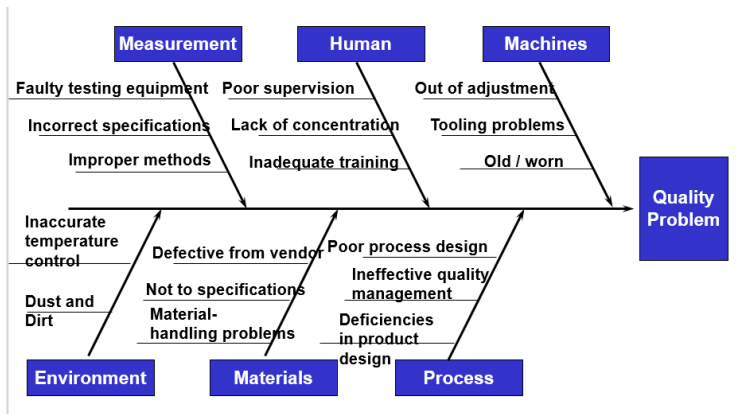
Pareto Chart



Control Chart



Fishbone Diagram



Statistical Process Control (SPC)

- ▶ Monitor processes over time.
- ▶ Detect unusual variation.
- ▶ Maintain stable production.
- ▶ Improve quality systematically.

Quality Characteristics

- ▶ Measurable variables describing fitness for use: weight, length, color, strength, or composition.
- ▶ Quality requires a clear **operational definition**.
- ▶ A target and tolerances determine conforming products.
- ▶ Example: 25 cm part with tolerance ± 0.01 cm; acceptable range [24.99, 25.01] cm.

Variation and Process Quality

- ▶ All processes exhibit variation.
- ▶ Statistical quality control measures and analyzes variation.
- ▶ Improve quality by centering the process on target and reducing variability.
- ▶ Reduce nonconforming products, sometimes measured in parts per million (PPM).

Sources of Variation

- ▶ **Common causes:** natural, inherent variation in a stable process.
- ▶ Reducing common-cause variation usually requires system improvement.
- ▶ **Special causes:** unusual events, equipment problems, or unexpected disturbances.
- ▶ Investigate and remove identifiable special causes.

Control Charts

- ▶ Track process performance over time to monitor stability.
- ▶ The center line represents the process average.
- ▶ Upper and lower control limits describe expected variation, often using three-sigma limits.
- ▶ Points outside the limits suggest potential special causes requiring investigation.

Process Stability and Control Charts

- ▶ A stable process has a quality-characteristic distribution that does not change over time.
- ▶ Control charts detect changes and unusual variation.
- ▶ Benefits include predictable production, early defect prevention, avoiding unnecessary adjustments, and diagnosing sources of variation.
- ▶ Control charts also inform process capability assessment.

Control Chart Implementation

- ▶ **Phase I:** collect baseline data, assess stability, and establish trial control limits.
- ▶ **Phase II:** periodically sample and monitor against established limits.
- ▶ Investigate signals of unusual process variation.
- ▶ Three-sigma limits are commonly used in practice.

Control Charts for Variables

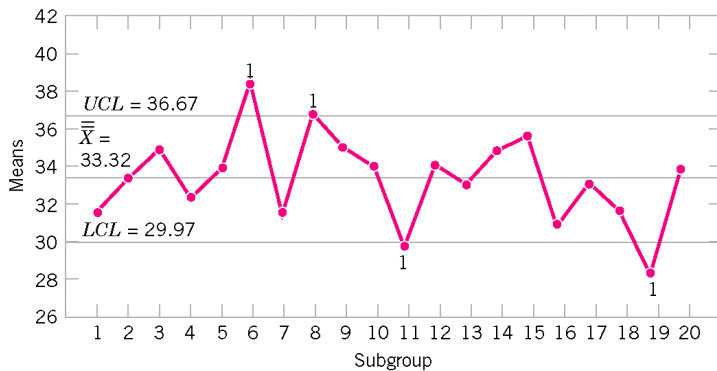
- ▶ The $\bar{X}$ -chart monitors the process mean using subgroup averages.
- ▶ The R -chart monitors short-term variation using within-subgroup ranges.
- ▶ Use both charts to assess stability of process location and variability.

Example: Vane-Opening Measurements

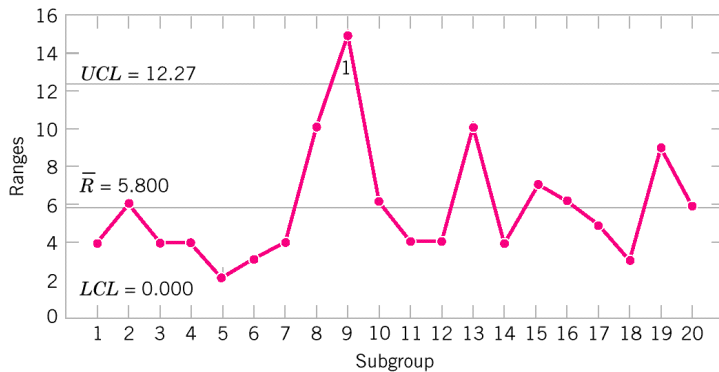
Table Vane-Opening Measurements

Sample Number	x_1	x_2	x_3	x_4	x_5	$\bar{x}$	r
1	33	29	31	32	33	31.6	4
2	33	31	35	37	31	33.4	6
3	35	37	33	34	36	35.0	4
4	30	31	33	34	33	32.2	4
5	33	34	35	33	34	33.8	2
6	38	37	39	40	38	38.4	3
7	30	31	32	34	31	31.6	4
8	29	39	38	39	39	36.8	10
9	28	33	35	36	43	35.0	15
10	38	33	32	35	32	34.0	6
11	28	30	28	32	31	29.8	4
12	31	35	35	35	34	34.0	4
13	27	32	34	35	37	33.0	10
14	33	33	35	37	36	34.8	4
15	35	37	32	35	39	35.6	7
16	33	33	27	31	30	30.8	6
17	35	34	34	30	32	33.0	5
18	32	33	30	30	33	31.6	3
19	25	27	34	27	28	28.2	9
20	35	35	36	33	30	33.8	6
						$\bar{\bar{x}} = 33.32$	$\bar{r} = 5.8$

$\bar{X}$ Chart for Vane Opening



R Chart for Vane Opening



Control Charts for Attributes

- ▶ Use attribute charts for counts or classifications.
- ▶ A p -chart monitors the proportion of defective units.
- ▶ A c -chart monitors the number of defects per item, based on a Poisson model.

p -Charts

- ▶ Based on the binomial distribution.
- ▶ Sample fraction defective: $\hat{p} = d/n$.
- ▶ Overall fraction defective:
 $\bar{p} = (\text{total defectives})/(\text{total inspected})$.
- ▶ Three-sigma control limits:

$$\text{UCL} = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}, \quad \text{LCL} = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n}}.$$

Example: Ceramic Substrates

Table Number of Defectives in Samples of 100 Ceramic Substrates

Sample	No. of Defectives	Sample	No. of Defectives
1	44	11	36
2	48	12	52
3	32	13	35
4	50	14	41
5	29	15	42
6	31	16	30
7	46	17	46
8	52	18	38
9	44	19	26
10	48	20	30

Example: p -Chart

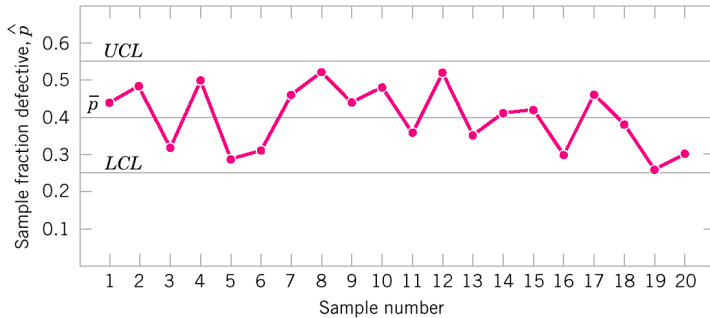


Figure 8-15 P chart for a ceramic substrate.

Control Charts for Individual Measurements

- ▶ Used when observations are collected one at a time ($n = 1$).
- ▶ Subgroup-based $\bar{X}$ - and R -charts cannot be used directly.
- ▶ Estimate variation using **moving ranges** between successive observations.
- ▶ Moving range: $MR_i = |X_i - X_{i-1}|$.

Individuals and Moving-Range Chart Limits

- Individuals chart:

$$UCL = \bar{\bar{x}} + 3\frac{\overline{MR}}{d_2}, \quad CL = \bar{\bar{x}}, \quad LCL = \bar{\bar{x}} - 3\frac{\overline{MR}}{d_2}.$$

- Moving-range chart:

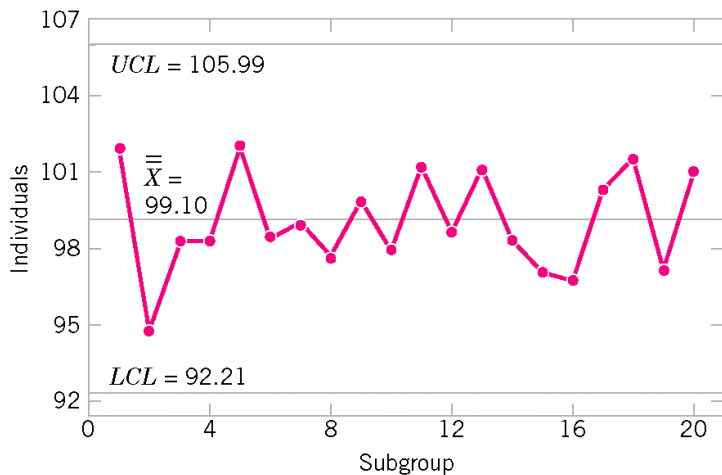
$$UCL = D_4\overline{MR}, \quad CL = \overline{MR}, \quad LCL = D_3\overline{MR}.$$

Example: Chemical Process Concentration

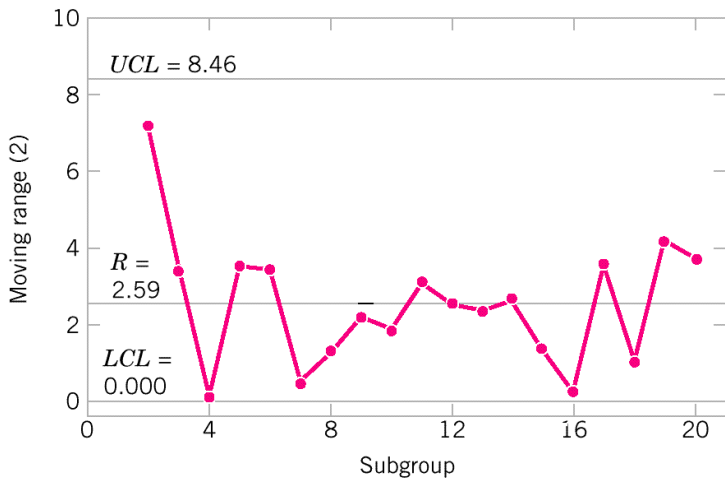
Table Chemical Process Concentration Measurements

Observation	Concentration x	Moving Range mr
1	102.0	
2	94.8	7.2
3	98.3	3.5
4	98.4	0.1
5	102.0	3.6
6	98.5	3.5
7	99.0	0.5
8	97.7	1.3
9	100.0	2.3
10	98.1	1.9
11	101.3	3.2
12	98.7	2.6
13	101.1	2.4
14	98.4	2.7
15	97.0	1.4
16	96.7	0.3
17	100.3	3.6
18	101.4	1.1
19	97.2	4.2
20	101.0	3.8

Example: Individuals Control Chart



Example: Moving-Range Control Chart



Process Capability

- ▶ Evaluate process capability **after establishing stability**.
- ▶ Determine whether products satisfy specification limits.
- ▶ Statistical control means a process is stable, not necessarily that products meet specifications.
- ▶ A capable process is centered near target with sufficiently small variation.

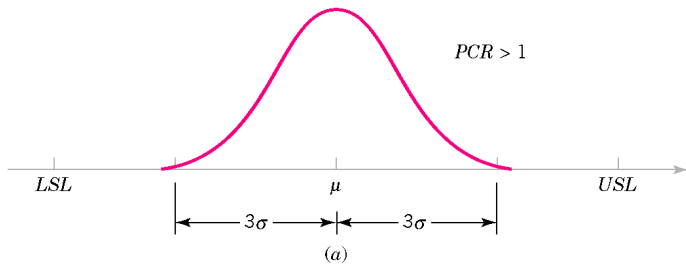
Process Capability Index

- ▶ Capability indices assess whether a stable process can meet specification limits.
- ▶ Common index:

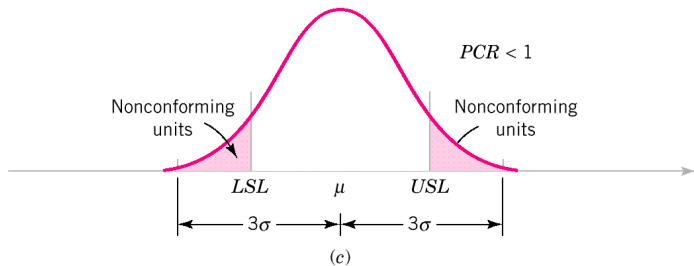
$$C_p = \frac{USL - LSL}{6s}.$$

- ▶ USL and LSL are specification limits; s estimates process standard deviation.
- ▶ $C_p > 1$ is a common minimum capability criterion; higher values may be required.

Process Capability Ratio



Process Capability Ratio



Key Takeaways

- ▶ SPC monitors and improves quality.
- ▶ Statistical experiments help improve products and processes.
- ▶ Variation is always present and requires statistical thinking.
- ▶ Control charts detect abnormal variation.
- ▶ SPC and quality tools support continuous improvement.

Quality of Artificial Intelligence Systems

- ▶ **New quality challenges.** AI systems introduce challenges for statistical quality control.
- ▶ **AI applications.** Examples include autonomous vehicles and large language models.
- ▶ **Sources of variation.** Performance depends on data, algorithms, and changing operating conditions.
- ▶ **Monitoring and improvement.** New methods are needed to evaluate and improve AI system reliability.

